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Genomic identification of the *NF-Y* gene family in apple and functional analysis of *MdNF-YB18* involved in flowering transition

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Abstract
Apple is a crucial economic product extensively cultivated worldwide. Its production and quality are closely related to the floral transition, which is regulated by intricate molecular and environmental factors. *Nuclear factor Y* (*NF-Y*) is a transcription factor that is involved in regulating plant growth and development, with certain *NF-Ys* play significant roles in regulating flowering. However, there is little information available regarding *NF-Ys* and their role in apple flowering development. In the present study, 51 NF-Y proteins were identified and classified into three subfamilies, including 11 MdNF-YAs, 26 MdNF-YBs, and 14 MdNF-YCs, according to their structural and phylogenetic features. Further functional analysis focused on *MdNF-YB18.* Overexpression of *MdNF-YB18* in *Arabidopsis* resulted in earlier flowering compared to the wild-type plants. Subcellular localization confirmed *MdNF-YB18* was located in the nuclear. Interaction between MdNFY-B18 and MdNF-YC3/7 was demonstrated through yeast two-hybrid (Y2H) and bimolecular fluorescence complementation (BiFC) assays. Yeast one-hybrid (Y1H) and the dual-luciferase reporter assays showed MdNF-YB18 could bind the promoter of *MdFT1* and activate its expression. Moreover, this activation was enhanced with the addition of MdNF-YC3 and MdNF-YC7. Additionally, MdNF-YB18 also could interact with *MdCOLs* (*CONSTANS Like*). This study lays the foundation for exploring the functional traits of MdNF-Y proteins, highlighting the crucial role of *MdNF-YB18* in activating *MdFT1* in *Malus*.

Keywords *Nuclear Factor Y (NF-Y)* · *Flowering Locus T (FT)* · *CONSTANS like* (*COL*) · Flowering transition · Apple

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Introduction

Nuclear Factor Y (*NF-Y*), also known as the CCAAT-binding factor (CBF), is a class of transcription factors widely found in eukaryotes. Gelinas et al. (1985) demonstrated that CCAAT-box is a *cis*-acting regulatory element that exists in the promoter regions of many genes (Bucher and Trifonov 1988). Initially discovered in yeast, proteins binding to the CCAAT-box function as part of the heme activator protein (HAP) complex, composed of four subunits: HAP2, HAP3, HAP4, and HAP5 (Forsburg and Guarente 1988). The *NF-Y* in plants is comprised of three subunits, including *NF-YA* (also called HAP2), *NF-YB* (also named HAP3) and *NF-YC* (also known as HAP5), and they can function individually or in the form of complexes (Mantovani 1999). *NF-Y* plays a crucial role in regulating gene expression by determining activation or repression in CCAAT-regulated genes (Donati et al. 2008). Previous studies have identified 36 *NF-Ys* in *Arabidopsis* (Siefers et al. 2009), 34 in rice (Yang et al. 2017), 50 in maize (Zhang et al. 2016), 25 in castor bean (Wang et al. 2018), 33 in walnut (Quan et al. 2018), 24 in peach (Li et al. 2019), 21 in peanut (Wan et al. 2021) and 23 in woodland strawberry (Zhou et al. 2024).

Plant growth and development, particularly the transition to flowering, are influenced by various factors (Kaur et al. 2021; Manzoor et al. 2023; Maple et al. 2024; Fan et al. 2024). *NF-Ys* play a crucial role in plant development (Mu et al. 2013; Bai et al. 2016), with a key impact on flowering regulation (Laloum et al. 2013). For example, *OsNF-YA8* is involved in vegetative and reproductive development in rice (Niu et al. 2023). *AtNF-YB2* and *AtNF-YB3* in *Arabidopsis* and *HvNF-YB1* (a homologous gene of *AtNF-YB2/3*) in barley have been reported to be participate in the flowering regulation network and promote *Arabidopsis* floral induction (Kumimoto 2008; Liang et al. 2012). *CONSTANS* (*CO*) serves as the central regulator in photoperiod pathways (Valverde 2004; Fornara et al. 2010), while *NF-YC3*, *NF-YC4*, and *NF-YC9* are known to regulate flowering either independently or through interaction with *CO* in *Arabidopsis* (Kumimoto et al. 2010). *PtCOL5-PtNF-YC4* is associated with reproductive cone development and gibberellin signaling in Chinese pine. *SiNF-YA3b* inhibits flowering in tomato by binding to the promoter of S*INGLE FLOWER TRUSS* (*SFT*) (Zhang et al. 2024). In citrus, *CiFT* (*Flowering locus T*) is activated by a complex composed of CiNF-YA1-CiNF-YB2-CiNF-YC2 (Zhou et al. 2022).

Flowering in plants is a complex program involving the transition from vegetative shoot meristem to a reproductive inflorescence, controlled by six main pathways focusing on the core flowering gene *FT*, *SUPPRESSOR OF OVEREXPRESSION OF CONSTANS1* (*SOC1*), *LEAFY* (*LFY*), *APETALA1* (*AP1*), and *FRUITFULL* (*FUL*) (Fornara et al. 2010). Among *FT* regulators, *CO* is the primary one whose protein stability is tightly controlled by phosphorylation and ubiquitination/proteasome-mediated mechanisms (Takagi et al. 2023). *CO* was identified to promote flowering and target *SOC1* and *FT* (Yoo 2005). The *NF-Ys* typically function as a trimer complex by binding to the distal enhancer element CCAAT-box on the *FT* promoter, facilitating CO to bind to the *FT* promoter and increasing its expression level (Cao et al. 2014). Additionally, another perspective suggests that CO regulates *FT* by displacing NF-YA to form a complex with the NF-YB/NF-YC dimer (Gnesutta et al. 2017). Additionally, NF-Y-mediated H3K27me3 demethylation regulates key flowering genes like *SOC1* and *LFY*, coordinating flowering responses in *Arabidopsis* (Liu et al. 2018; Hou et al. 2014; Winter et al. 2011). This regulatory pathway also integrates genes such as DELLA protein in gibberellin and miRNA156 in age pathways, influencing the flowering process (Hou et al. 2014; Bao et al. 2020).

Apple (*Malus* x *domestic*a Borkh.) is a prominent temperate fruit tree in the Rosaceae family, extensively grown in various countries including China. Understanding the mechanisms behind apple flowering induction can help manage flowering time and enhance production. *MdFT1* has been identified as the key factor activating apple flowering. *MdbHLH4* has been shown to promote apple flowering by interacting with *MdFT* (Wang et al. 2023). Additionally, other apple transcription factors such as *MdGRF* and *MdBLH4* have been reported to be involved in regulating flowering (Zuo et al. 2022; Jia et al. 2023). But there is limited information about NF-Y family proteins in apples and it is unclear whether NF-Ys can regulate *MdFT1* by forming complexes similar to those in *Arabidopsis*. In this study, 51 *MdNF-Ys* were identified, and their gene structures, chromosomal locations, and evolutionary relationships were analyzed. Expression patterns were examined to understand their responses across different periods and varieties. We investigated the role of *MdNF-YB18* in regulating the flowering transition in both *Arabidopsis* and *Malus*. This study will provide valuable insights into apple flowering transition and provide a foundation for enhancing apple yield.

Material and methods

Plant materials and growth conditions

Apple trees, including 'Nagafu No.2' and 'Qinguan', were grown under natural conditions in Yangling, Shaanxi, China. Buds were collected 30 days after the full bloom, a period considered crucial for floral induction. For the photoperiodic response analysis, buds were harvested every four hours.

The wild-type and transgenic *Arabidopsis* seedings are of the Columbia type (Col-0). Sterilized seeds were cold-stratified for 2 days, germinated on medium plates, and then transplanted in soil mixed with vermiculite. The plants were grown under long days (16 h of light and 8 h of dark) until flowering. The number of rosette leaves was measured during the bolting period, and data from 28 individual plants were collected and analyzed statistically.

Identification and chromosomal location of *NF-Y* genes in apple

A total of 36 full-length NF-Y protein sequences from *Arabidopsis* (http://www.arabidopsis.org/) were used as queries in BLASTp search (E-value < 1e-5) against the *Malus* x *domestica* GDDH13 v1.1 genome available in the GDR database (https://www.rosaceae.org/blast/protein/protein) to identify candidate *MdNF-Y* sequences. All candidate NF-Y sequences of *Malus domestica* were submitted to the NCBI Conserved Domains Database (CDD, https://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi) to verify the presence of conserved domains. The chromosomal distribution of putative *MdNF-Ys* was analyzed using the *Malus domestica* Genome Database. All selected *MdNF-Ys* were classified into three subfamilies and renamed based on their chromosomal locations.

Gene structure, *cis*-elements, synteny analysis, and phylogenetic tree construction analysis

The software TBtools (Chen et al. 2020) was used to extract information on the 51 *MdNF-Y* genes from the apple genome, generating the CDS (coding sequence)-UTR (untranslated regions) structures and identifying conserved motifs. Full-length protein sequences were submitted to the Sopma Secondary Structure Prediction website (https://npsa-prabi.ibcp.fr/cgi-bin/npsa_automat.pl?page=npsa_sopma.html) and Phyre2 (http://www.sbg.bio.ic.ac.uk/phyre2/html/page.cgi?id=index) for secondary and tertiary structure predictions. The characterization of NF-Y proteins in apple was performed using ExPASy (https://web.expasy.org/protparam/). To identify potential *cis*-elements in promoter regions, the 3000 bp intergenic regions upstream of of the *MdNF-Y* start codons were extracted from the apple genome and analyzed using the PlantCARE website (Lescot et al. 2002). Chromosome distribution and synteny analysis of *NF-Ys* in apple and *Arabidopsis* were visualized using circus version 0.63. For phylogenetic tree construction, the Muscle software was used to align the protein sequences, and the MEGA IX software was used to build the tree using the Neighbor-Joining method with 1000 bootstrap replicates.

Expression patterns of MdNF-Ys in different varieties and organs

We compared and investigated the relative expressions of *MdNF-Y*s in early stage (ES), middle stage (MS) and late stage (LS) of buds after full bloom, which obtained from RNA-sequence data of 'Nagafu No.2' (the classical variety in difficult-to-flowering) and 'Qinguan' (the classical variety in easy-to-flowering) (Xing et al. 2016).

To better understand gene expression patterns, we downloaded expression data from various organs (flower, young fruit, leaf, fruit, root, stem, seedling, and seed) across ten apple cultivar genotypes from Array Express Data (E-GEOD-GSE42873) and extracted information for 51 *MdNF-Ys* genes. Heat maps of *MdNF-Ys* gene expression were generated using Heml 1.0 software.

Plant transformation and flowering time experiments

Arabidopsis thaliana ecotype Columbia (Col-0) served as the wild type for this experiment. The full-length CDS sequence of *MdNF-YB18* was cloned into the PRI101 vector. Plants were transformed using Agrobacterium-mediated floral dipping and then grown in an incubator-like room under standard long-day conditions (16 h light/8 h dark), consistent with previously reported growth conditions (Narusaka et al. 2010). All transgenic plants were selected on medium containing kanamycin (50 μg/ml). The total number of rosette leaves was recorded during flowering.

Yeast two-hybrid analysis

The full-length sequence of *MdNF-YB18* was cloned into the bait vector pGBKT7 and tested for self-activation and self-toxicity in the yeast two-hybrid assay. The cDNA library from flower buds was utilized to identify potential proteins interacting with MdNF-YB18. All plaques on the selective medium (QDO: SD medium lacking Leu, Trp, His, and Ade) were sequenced. Following the Matchmaker Gold Yeast two-Hybrid System (Clontech), the proteins MdNF-YC3, MdNF-YC7, MdCOL2a, MdCOL2b, MdCOL4a, MdCOL4b, MdCOL5, and MdCOL12 were cloned into the pGADT7 vector. Interaction between bait and prey proteins was assessed on QDO medium with or without X-a-Gal.

Subcellular location

The full-length coding sequence of *MdNF-YB18*, *MdNF-YC3*, and *MdNF-YC7* excluding the stop codon, were cloned into the pCAMBIA2300-GFP vector, respectively. The specific methods refer to the previous study (Zuo et al. 2021).

Bimolecular fluorescence complementation assays

For BiFC assays, the full-length coding sequences of *MdNF-YB18, MdNF-YC3*, and *MdNF-YC7* were fused in-frame with the pSPYCE and pSPYNE vectors, respectively. *Agrobacterium tumefaciens* GV3101 was used, following the methods described in a previous study (Zuo et al. 2021).

Yeast one-hybrid analysis

The promoter of *MdFT1* was examined for the yeast one-hybrid assay, revealing the presence of several CCAAT-box elements. Notably, a sequence comprising three consecutive CCAAT-box elements was identified in the region spanning from −950 bp to −150 bp. This fragment, named pFT1-3, was inserted into the pAbAi vector and then transformed into the Y1H Gold strain after linearization. Additionally, MdNF-YB18 and six MdCOLs were cloned into a pGADT7 vector, respectively. The interaction between *pFT1-3*-pAbAi and these proteins expressed in the pGADT7 vectors was assessed using SD/-Leu medium (synthetic dextrose lacking leucine) supplemented with AbA (Aureobasidin A).

Dual-luciferase assay

The 1500-bp intergenic regions upstream of the start codon of *MdFT1* were constructed into vector pGreenII 0800-LUC, and the full-length coding sequences of *MdNF-YB18*, *MdNF-YC3*, and *MdNF-YC7* were constructed into vector pGreenII 62-SK, respectively. The transformation was carried out by inserting each set of vectors into *Agrobacterium tumefaciens* GV3101 (with pSoup), and the constructs were introduced into the epidermal cells of *N. benthamiana* plants that were 3–4 weeks old. After 2 days of incubation in darkness, the infected region of the leaf was cut, smashed, and gathered into tubes and stored in −80 ℃ conditions. The smash samples were treated according to the Dual-Luciferase® Reporter Assay System of Promega, and the relative values of LUC/REN were tested in three replicates.

Results

Identification and genomic localization of *NF-Ys* in apple

By removing incomplete and redundant sequences, 51 *MdNF-Ys* were identified and classified into three subfamilies, 11 *MdNF-YAs*, 26 *MdNF-YBs,* and 14 *MdNF-YCs* (Table 1). These 51 *MdNF-Ys* were renamed based on their distribution and relative position on apple chromosomes (Fig. S1) and spread across 16 chromosomes, excluding Chr8. The number of genes per chromosome ranges from 1 (Chr1 and Chr17) to 5 (Chr2, Chr3, and Chr11). *MdNF-YBs* are found on all chromosomes except Chr 8 and Chr16. In contrast to the distribution of *NF-Ys* in tomatoes reported by Li et al. (2016), 4 *MdNF-YBs* were respectively identified on Chr 3 and Chr11, each containing the highest number of *NF-YBs*. This suggests that the *NF-Y* gene distribution varies among species and is uneven tacross plant genomes.

The length of the 51 MdNF-Y proteins ranged from 71 to 358 amino acids, with MdNF-YB6 being the shortest and MdNF-YA3 the longest (Table 1). Most MdNF-YAs had a molecular weight above 30 kDa, except for MdNF-YA4 (15.60 kDa) and MdNF-YA7 (22.82 kDa). Within subfamily B, protein weights ranged from 8.14 kDa (MdNF-YB6) to 38.13 KDa (MdNF-YB3), while those in subfamily C averaged around 26 KDa. The instability of proteins in subfamilies A and C was evident with an isoelectric point above 40, whereas MdNF-YBs were found to be more stable according to Table 1.

Structure analysis of *MdNF-Ys*

In the phylogenetic tree, the gene structures of the *NF-Y* family in apples were highly similar within each subfamily. We analyzed the UTR-CDS structures and conserved motifs of the 51 *MdNF-Ys* based on their genome sequences (Fig. 1). All *MdNF-YA* genes have two conserved motifs (motif 3 and motif 5), while most *MdNF-YB* and *MdNF-YC* genes have three and two conserved motifs, respectively. All *MdNF-YAs* contain introns, whereas only nine *MdNF-YBs* (*MdNF-YB1, 8, 11,*

Table 1 Physical and chemical characters of amino acid sequences among *MdNF-Ys*

Gene Name	Gene ID	Location	Strand	CDS (bp)	Peptide (aa)	MW[b] (KDa)	PI[d]	II[e]	AI[f]	GRAVY[c]
NF-YAs										
MdNF-YA1	MD02G1086200	Chr2: 6,760,401–6763923	-	927	308	33.93	9.28	50.6	58.21	−0.699
MdNF-YA2	MD02G1309800	Chr2: 36,538,789–36,544,223	-	885	294	33.54	8.87	51.59	46.12	−1.004
MdNF-YA3	MD03G1174600	Chr3: 23,867,258–23,872,067	-	1077	358	38.86	6.73	57.87	45.08	−0.994
MdNF-YA4	MD05G1273300	Chr5: 40,831,929–40836805	+	414	137	15.6	9.93	74.6	63.43	−1.088
MdNF-YA5	MD07G1011300	Chr7: 1,023,634–40836805	+	840	279	30.58	8.69	50.38	53.15	−0.905
MdNF-YA6	MD09G1186200	Chr9: 16,163,341–16,168,480	+	897	298	32.86	9.44	52.81	69.66	−0.619
MdNF-YA7	MD10G1253300	Chr10: 34,584,154–34,588,714	+	627	208	22.82	7.91	65.28	54.9	−0.963
MdNF-YA8	MD11G1192400	Chr11: 27,332,752–27,339,954	-	1047	348	37.92	8.66	57.57	43.28	−1.109
MdNF-YA9	MD12G1042100	Chr12: 4,650,199–4655169	-	969	322	35.11	9.21	44.84	58.79	−0.622
MdNF-YA10	MD14G1041300	Chr14: 3,873,166–3,878,129	-	930	309	33.94	9.64	47.6	54.66	−0.679
MdNF-YA11	MD15G1213400	Chr15: 17,122,823–17,125,801	-	903	300	33.21	9.26	44.29	66.27	−0.589
NF-YBs										
MdNF-YB1	MD01G1112400	Chr1: 22,658,861–22,662,399	-	645	214	22.93	8.78	49.39	70.7	−0.507
MdNF-YB2	MD02G1191900	Chr2: 17,939,911–17,940,645	-	735	244	27.02	5.27	64.68	47.17	−0.776
MdNF-YB3	MD02G1192100	Chr2: 18,023,631–18025409	-	1044	347	38.13	4.89	60.08	66.34	−0.482
MdNF-YB4	MD03G1179600	Chr3: 24,691,097–24691663	+	567	188	20.69	4.88	58.37	63.35	−0.428
MdNF-YB5	MD03G1183400	Chr3: 25,025,010–25025594	-	585	194	20.84	6.44	37.25	52.27	−0.837
MdNF-YB6	MD03G1280100	Chr3: 36,149,310–36149647	+	216	71	8.14	4.83	51.15	70	−0.151
MdNF-YB7	MD03G1283700	Chr3: 36,413,956–36,414,516	-	561	186	20.95	6.66	42.84	58.28	−0.895
MdNF-YB8	MD04G1104700	Chr4: 19,216,567–19,220,090	-	525	174	18.78	6.61	46.05	55	−0.884
MdNF-YB9	MD04G1203300	Chr4: 28,953,974–28,954,426	+	453	150	16.86	6.3	34.43	67.73	−0.749
MdNF-YB10	MD05G1361000	Chr5: 47,685,520–47685846	+	327	108	11.91	5.21	41.2	67.78	−0.694
MdNF-YB11	MD05G1361600	Chr5: 47,696,633–47,698,977	+	876	291	31.27	6.63	31.72	56.12	−0.627
MdNF-YB12	MD06G1209300	Chr6: 34,273,320–34276783	+	471	156	17.52	4.72	52.74	76.35	−0.537

Table 1 (continued)

Gene Name	Gene ID	Location	Strand	CDS (bp)	Peptide (aa)	MW^b (KDa)	PI^d	II^e	AI^f	$GRAVY^c$
MdNF-YB13	MD07G1180200	Chr7: 25,950,191–25953495	-	648	215	23.16	8.8	50.45	70.37	−0.541
MdNF-YB14	MD09G1126900	Chr9: 9,795,758–25,953,495	+	654	217	22.18	7.02	43.42	47.24	−0.735
MdNF-YB15	MD10G1339000	Chr10: 41,496,215–41,496,999	+	744	247	27.35	7.1	41.76	56.19	−0.601
MdNF-YB16	MD11G1164600	Chr11: 16,978,966–16,980,190	+	501	166	18.19	8.73	49.92	66.45	−1.027
MdNF-YB17	MD11G1199300	Chr11: 28,769,434–28,769,985	+	552	183	20.25	5.59	55.8	47.49	−0.667
MdNF-YB18	MD11G1200400	Chr11: 28,876,837–28,877,409	-	573	190	20.09	6.91	31.41	55.95	−0.684
MdNF-YB19	MD11G1302500	Chr11: 41,788,083–41788580	+	498	165	18.6	6.38	30.75	61.52	−0.92
MdNF-YB20	MD12G1217100	Chr12: 29,460,289–29460741	+	453	150	16.79	7.69	30.65	69.67	−0.727
MdNF-YB21	MD12G1124800	Chr12: 19,970,949–19980773	-	393	130	14.28	5.82	31.11	59.31	−0.874
MdNF-YB22	MD13G1170200	Chr13: 13,815,353–13,815,826	+	474	157	17.51	6.31	36.37	63.44	−0.91
MdNF-YB23	MD14G1219800	Chr14: 30,211,443–13815826	+	471	156	17.51	4.72	53.78	77.56	−0.519
MdNF-YB24	MD15G1134600	Chr15: 9,778,391–9,779,101	+	711	236	26.72	6.52	50.44	57.42	−1.01
MdNF-YB25	MD15G1334200	Chr15: 37,152,321–37,152,947	+	627	208	23.01	7.09	41.87	54.42	−0.86
MdNF-YB26	MD17G1117200	Chr17: 10,167,264–10168240	+	720	239	24.81	5.48	43.66	52.22	−0.65
NF-YCs										
MdNF-YC1	MD02G1273400	Chr2: 32,804,138–32809135	+	999	332	36.01	5.11	48.87	68.04	−0.704
MdNF-YC2	MD04G1207500	Chr4: 29,269,117–29,271,406	-	807	268	29.75	6.16	60.07	69.59	−0.578
MdNF-YC3	MD05G1229600	Chr5: 36,298,931–36,300,862	+	807	268	29.72	5.33	69.18	73.58	−0.478
MdNF-YC4	MD06G1078500	Chr6: 19,362,575–19,362,940	+	366	121	13.37	7.67	61.28	81.32	−0.161
MdNF-YC5	MD06G1141200	Chr6: 28,495,891–28,498,876	+	720	239	25.83	5.19	64.05	71.13	−0.409
MdNF-YC6	MD07G1042300	Chr7: 3,543,314–3,555,874	+	978	325	34.99	5.05	43.42	69.54	−0.683
MdNF-YC7	MD10G1208800	Chr10: 30,806,582–30809166	+	834	277	31.17	5.84	70.04	66.93	−0.636
MdNF-YC8	MD12G1221800	Chr12: 29,861,745–29,864,137	-	789	262	29.22	5.8	61.59	63.36	−0.617
MdNF-YC9	MD13G1118300	Chr13: 8,645,120–8645611	-	492	163	17.99	5.58	64.07	77.91	−0.564

Table 1 (continued)

Gene Name	Gene ID	Location	Strand	CDS (bp)	Peptide (aa)	MW[b] (KDa)	PI[d]	II[e]	AI[f]	GRAVY[c]
MdNF-YC10	MD13G1185200	Chr13: 15,766,106–15768771	+	756	251	28.01	5.91	52.49	68.49	−0.643
MdNF-YC11	MD14G1156400	Chr14: 25,090,589–25093134	+	717	238	25.78	5.05	64.77	69.79	−0.405
MdNF-YC12	MD15G1159400	Chr15: 11,916,470–11917171	-	702	233	26.52	5.35	59.54	74.59	−0.157
MdNF-YC13	MD16G1118400	Chr16: 8,414,848–11,917,171	-	585	194	21.6	4.81	72.46	87.06	−0.589
MdNF-YC14	MD16G1185900	Chr16: 16,034,439–16037111	+	699	232	26.06	5.9	62.78	70.3	−0.673

MW: Molecular weight, *PI*: Theoretical pI, II: Instability index, *AI*: Aliphatic index, *GRAVY*: Grand average of hydropathicity

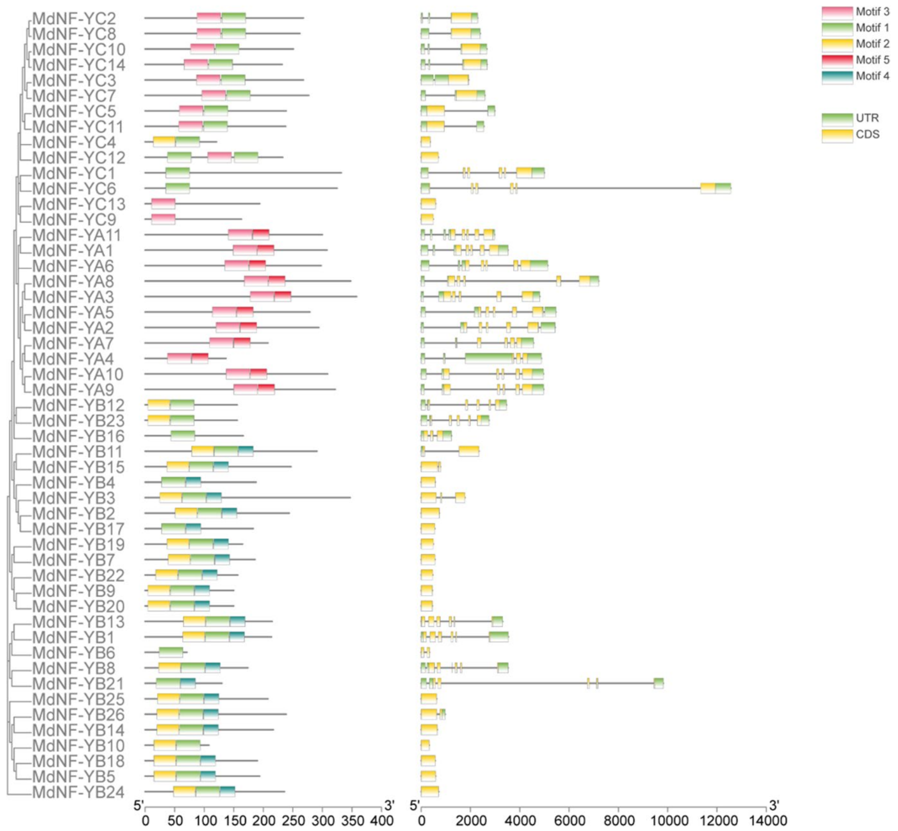

Fig. 1 Motif analysis and CDS/UTR organization of *MdNF-Ys* genes. The five main motifs are marked in pink to blue. In Exon–intron composition, The green and yellow boxes, and black lines, represent UTR, CDS, and intron positions, respectively

12, 13, 16, 21, 23, and *26*) include UTR regions, and four *MdNF-YCs* (*MdNF-YC4, 9, 12,* and *13*) involve only CDS regions. *MdNF-YAs* exhibit complex and variable introns among different members, consistent with the previous research (Ripodas et al. 2014).

To gain a better understanding of the protein structure of *MdNF-Ys*, both the secondary and tertiary structures were predicted and analyzed. The tertiary structure is formed through the winding and folding of the secondary structure, with spatial configuration maintained mainly by hydrophobic hydrogen bonds among amino acid chains. The secondary structure comprises alpha helices, beta turns, extended strands, and random coils. In all three subfamilies, alpha helices and random coils were most prevalent. However, in MdNF-YAs, the percentage of random coils is significantly higher than that of alpha helices. A more balanced distribution was observed in subfamilies B and C (Table. S1). These findings align with those reported by Quach et al. (2015). Furthermore, the 51 proteins were categorized into three distinct spatial structures corresponding to each subfamily (Fig. S2), indicating significant conservation within this protein family.

Phylogenetic relationships and synteny analysis of NF-Y proteins

To investigate the evolutionary relationships and functional divergence of the *MdNF-Ys*, we constructed a phylogenetic tree using full-length protein sequences of MdNF-Ys and AtNF-Ys with the neighbor-joining method in MEGA IX (Fig. 2a). The sequences were classified into three subgroups (NF-YA, NF-YB, and NF-YC) based on the phylogenetic tree (Saitou and Nei 1987). We used the Circos program to assess segmental and tandem duplications of MdNF-Y genes. Many MdNF-Y gene pairs, such as *MdNF-YB14/26*, *MdNF-YC9/14*, *MdNF-YB1/13*, and *MdNF-YA1/6*, were found in duplicated genomic regions (Fig. 2b). These duplicated regions are on different chromosomes, and the presence of these gene pairs can partly be attributed to chromosome replication in apples.

Cis-elements analysis in the promoters of *MdNF-Ys*

Characterizating *cis*-acting elements can reveal insights into gene function and protein interactions. To explore the roles of *MdNF-Ys* under various conditions, we analyzed the *cis*-acting elements of 51 *MdNF-Ys* (Fig. 3). The CAAT-box is the most prevalent element in the promoter regions, making it the most common type. Additionally, the promoter regions of *MdNF-Ys* include elements responsive to hormones (ABRE, P-box, TGA-element, TCA-element), light (G-Box), stress (W-box, MBS), plant growth (circadian), and other pathways (F-box). *Cis*-acting elements in subfamilies A and C are relatively dispersed, while subfamily B shows a higher concentration and greater number of these elements. Overall, these results suggest that *MdNF-Y* genes may be involved in various aspects of plant growth, development, and hormone response.

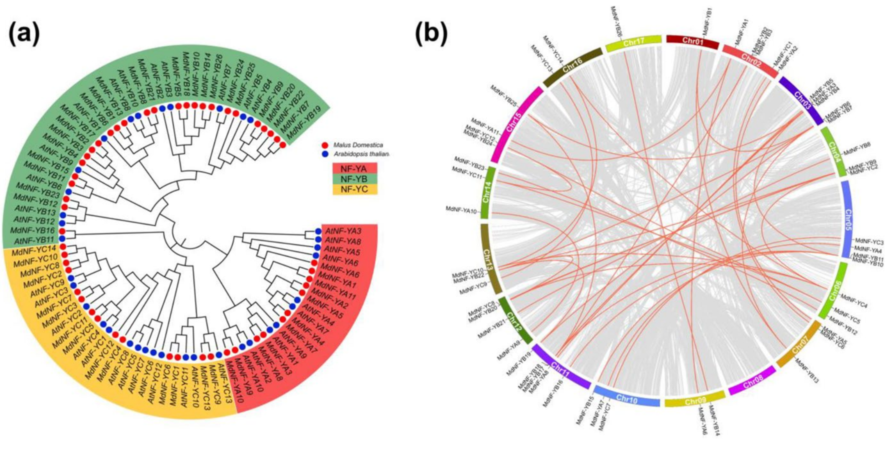

Fig. 2 Evolutionary relationship of NF-Y gene family. (**a**). Phylogenetic tree showing the evolutionary relationships of MdNF-Y proteins in apple and *Arabidopsis*. Protein designations consist of the prefixes *Malus domestica* (Md, red triangles) and *Arabidopsis thaliana* (At, blue circles). The three subfamilies of NF-Y proteins were marked in red (NF-YA), green (NF-YB) and yellow (NF-YC). (**b**). Chromosomal distribution and duplication relationships of *NF-Y* genes in apple

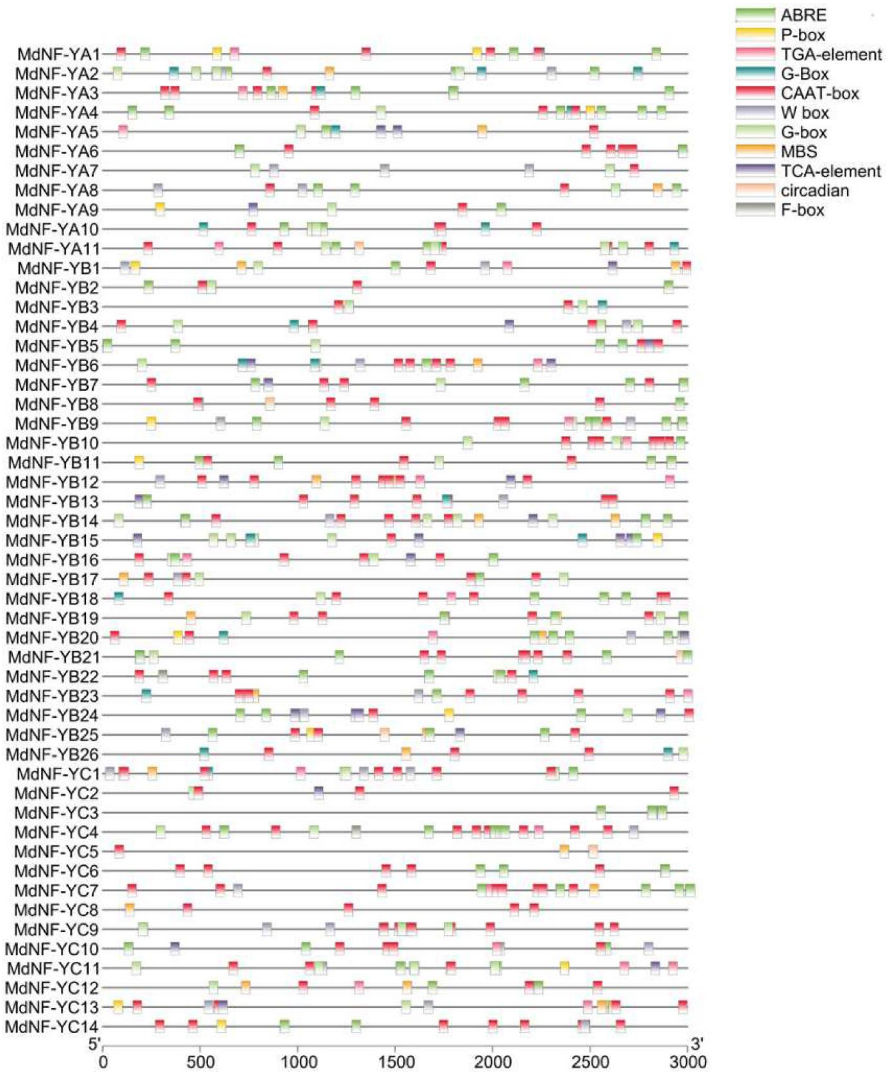

Fig. 3 Analysis of the *cis-elements* in the *MdNF-Y* promoters. The diverse elements were represented by boxes in different colors

Expression patterns of *MdNF-Ys* in different organs and varieties

The expression patterns of *MdNF-Ys* were analyzed in two cultivated varieties: the easily flowering variety ‘QinGuan’ and the difficult-to-flower variety ‘Nagafu No.2’. Additionally, expression profiles were examined across ten different varieties using data from the ArrayExpress Database (E-GEOD-GSE42873). Expression levels of subfamilies A and B varied significantly between the two varieties in buds (Fig. 4a). Notably,

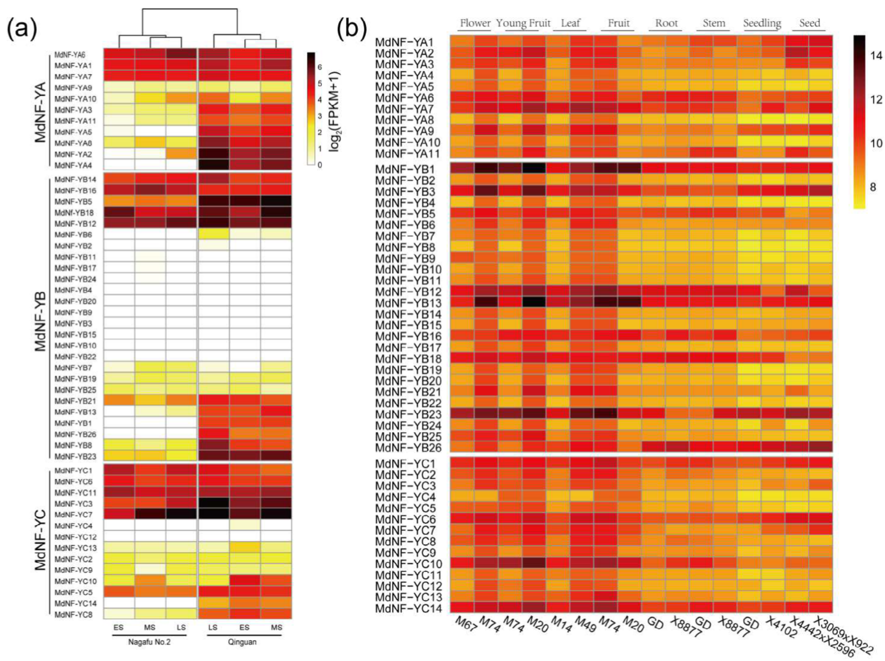

Fig. 4 Expression profiles of *MdNF-Ys*. (**a**). Expression patterns from RNA-Seq data in 'Nagafu No.2' and 'Qinguan'. 'Nagafu No.2' is a variety difficult to blossom and 'Qinguan' is that easy to blossom. ES, MS and LS represent the early, middle and late stages of buds after full blossom, respectively. (**b**). Expression patterns of *MdNF-Ys* in different tissues in virous varieties

the expression of *MdNF-YB18* was much higher in 'Qinguan' compared to 'Nagafu No.2'. Among the three subfamilies, genes in the *MdNF-YB* and *MdNF-YC* subgroups were highly expressed (Fig. 4b). Specifically, members such as *MdNF-YB1*, *3*, *12*, *13*, *18*, and *23*, as well as *MdNF-YC10* and *MdNF-YC14*, showed relatively higher expression levels in flower tissues. These results suggest that genes involved in the flowering transition in *MdNF-Ys* are primarily concentrated in subfamilies B and C.

Overexpression of *MdNF-YB18* accelerates flowering in *Arabidopsis*

RT-qPCR analysis revealed that *MdNF-YB18* and *MdFT1* exhibit similar expression patterns across various growth stages in 'Qinguan' and 'Nagafu No.2' (Fig. 5a) and under varying photoperiod conditions (Fig. 5b). Both genes showed increased expression during the middle stage (MS) of flower bud differentiation, peaking at 10:00 a.m.

To investigate the role of *MdNF-YB18* in regulating flowering, we created *Arabidopsis* lines with its overexpression. These transgentic lines exhibited earlier flowering and a significant reduction in rosette leaves after bolting compared to the wild type (Fig. 5c and d), indicating that *MdNF-YB18* likely promotes flowering in *Arabidopsis*.

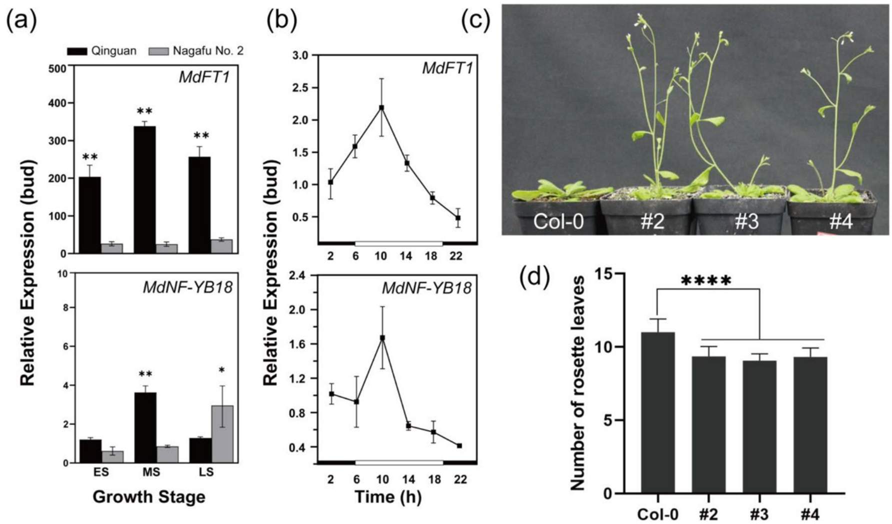

Fig. 5 *MdNF-YB18* accelerate the flowering in *Arabidopsis*. (**a**). Expression patterns of *MdFT1* and *MdNF-YB18* in different growth stage between two varieties. (**b**). Expression patterns of *MdFT1* and *MdNF-YB18* under photoperiod. (**c**). Phenotype of *MdNF-YB18* overexpressed lines. D. Numbers of rosette leaves of *MdNF-YB18* overexpressed lines. (*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$, ****$P < 0.0001$, Student's *t*-test)

MdNF-YB18 interacts with MdNF-YC3 and MdNF-YC7

Since NF-Ys function by forming dimer or trimer complex, we used MdNF-YB18 as bait in a yeast two-hybrid screen to identify interacting proteins. This screening revealed potential candidates, including MdNF-YC3 and MdNF-YC7. The yeast two-hybrid assay confirmed that MdNF-YB18 interacts with both MdNF-YC3 and MdNF-YC7 (Fig. 6a). To pinpoint the functional sites of MdNF-YB18, MdNF-YC3, and MdNF-YC7, we analyzed their gene sequences using the NCBI conserved domain database and truncated the genes based on conserved domain locations. The results show that only the truncated fragment of MdNF-YB18, containing the CBF domain (Fig. S3), interacts with MdNF-YC3 and MdNF-YC7. Similarly, MdNF-YC3 and MdNF-YC7 also operate through the HAP5 conserved domain (Fig. S4).

We also investigated subcellular localization of MdNF-YB18, MdNF-YC3 and MdNF-YC7. The fluorescent signals observed in the localization assay were present in nucleus (Fig. 6b), confirming that these proteins are nuclear transcription factors. In a bimolecular fluorescence complementation (BiFC) assay (Fig. 6c), MdNF-YB18 was fused with the split C-terminal of Yellow Fluorescent Protein (YFP) (pSYCE), while MdNF-YC3 and MdNF-YC7 was fused with the split N-terminal of YFP (pSYNE), respectively. Strong YFP signals was detected when we transiently co-expressed MdNF-YB18-CE and MdNF-YC3-NE or MdNF-YC7-NE in *Nicotiana benthamiana* leaves. However, no detectable YFP signal was seen in the negative controls. These results indicated that MdNF-YB18 interacts with both MdNF-YC3 and MdNF-YC7, suggesting its role as part of a dimeric complex involved in floral induction regulation.

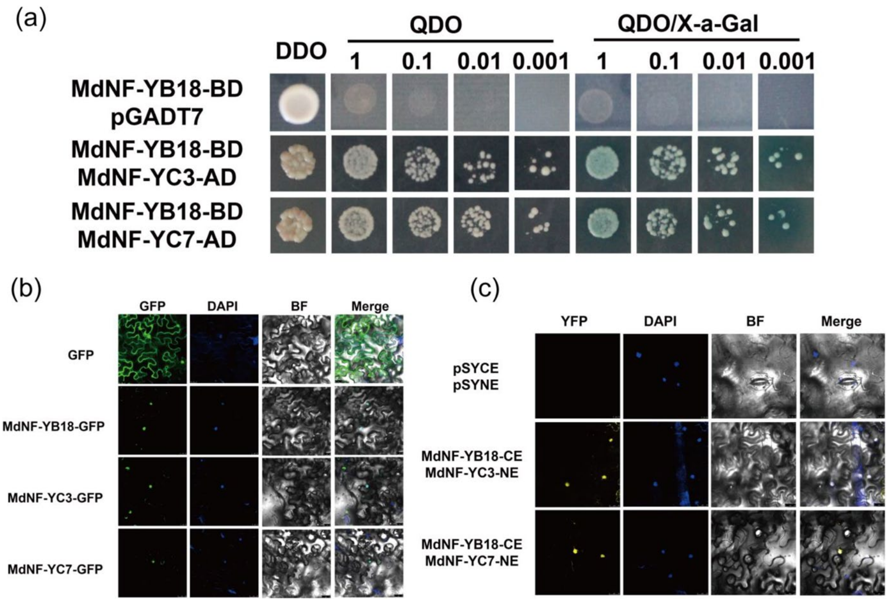

Fig. 6 Subcellular localization and interaction between MdNF-YB18 and MdNF-YC proteins. (**a**). Yeast two-hybrid assays. The empty pGADT7 vector was used as control. DDO, yeast medium lacking leucine and tryptophan. QDO, yeast medium lacking leucine, tryptophan, adenine and histidine. (**b**). Subcellular localization. The empty PC2300 vector fused with green fluorescent protein (GFP) coding region was used as the control. DAPI (4,6-diamidino-2-phenylindole) was used for the nucleus staining. The GFP fluorescence, DAPI, and bright-field images were merged. Samples were imaged by laser scanning confocal microscopy (LEICA TCS SP8, Germany). Bar = 25 μm. (**c**). BiFC assays. The empty pSPYCE and pSPYNE vector served as the blank and negative control. Fluorescence was imaged by laser scanning confocal microscopy (LEICA TCS SP8, Germany). The YFP fluorescence, DAPI, and bright-field images were merged. Bar = 25 μm

MdNF-YB18 promotes *MdFT1* expression by interacting with MdNF-YC3 and MdNF-YC7

CO and NF-Y are known to compete in regulating the expression of *FT* in *Arabidopsis* (Wenkel et al. 2006; Zhao et al. 2016), with CO consistently playing a central role in photoperiod pathways (Valverde 2004; Fornara et al. 2010). In this study, In this study, we identified *MdCO-Likes* (*MdCOLs*) in apple (Fig. S5) and constructed them into pGADT7 vectors. The Y2H assay demonstrated that all selected MdCOLs (MdCOL2a, MdCOL2b, MdCOL4a, MdCOL4b, MdCOL5, and MdCOL12) can interact with MdNF-YB18 (Fig. 7a).

The promoter of *MdFT1* contains three consecutive CCAAT boxes, which are known binding elements for NF-YB. To determine if MdNF-YB18 activates *MdFT1* expression by binding to these CCAAT boxes, a Y1H assay was performed. The results showed that MdNF-YB18 binds to the *pFT1-3* fragment containing the three CCAAT boxes (Fig. 7b). Additionally, we tested whether MdCOLs could activate *MdFT1-3* expression, but they do not activate it as MdNF-YB18 does.

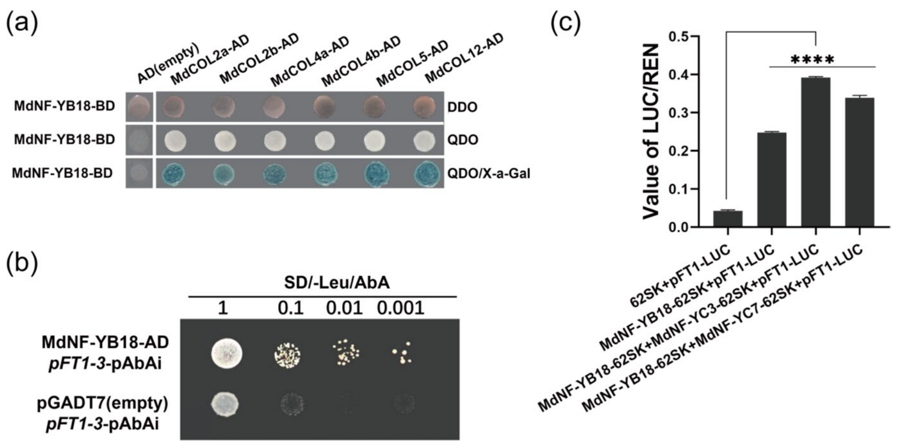

Fig. 7 MdNF-YB18 activated the expression of *MdFT1* by interacted with MdNF-YC3/7. (**a**). The interaction between MdNF-YB18 and MdCOLs. (**b**). MdNF-YB18 promote the expression of *MdFT1* by binding to the promoter. (**c**). The interaction between MdNF-YB18 and MdNF-YC3/7 activate the expression of *MdFT1* promoter

To investigate whether MdNF-YB18 and MdNF-YCs interact to influence *MdFT1*, we performed dual-LUC experiments (Fig. 7c). After introducing MdNF-YB18, the pFT-LUC activity value increased to eight times that of the negative control. Moreover, adding either MdNF-YC3 or MdNF-YC7 further boosted the pFT-LUC activity.

Discussion

The apple yield is significantly affected by flowering. Understanding the regulatory mechanisms of apple flowering can boost yields. NF-Y proteins are present in most eukaryotes, indicating their potential functional complexity. Research shows that *NF-Y* plays a role in plant flowering, investigating this gene family can provide insights into apple flowering mechanisms.

This study utilized 36 AtNF-Y proteins as queries and identified 51 NF-Y members in apple, examining their structural and protein characteristics. The 51*MdNF-Ys* were categorized into three subfamilies based on evolutionary relationships and conserved regions (Figs. 1 and 2). Essential regulatory roles in plants are played by transcription factors such as MYB (Liu et al. 2024), WRKY (Yu et al. 2023), and WOX (Khan et al. 2024). Numerous studies have highlighted the involvement of NF-Y proteins, either individually or collectively, in drought stress resistance (Chen et al. 2014; Lee et al. 2015), floral induction (Cao et al. 2014; Hou et al. 2014; Wei et al. 2017), and plant development (Battaglia et al. 2014; Zanetti et al. 2017). Analysis of promoter has revealed that these family members possess diverse functional roles due to the abundance of cis-acting elements within their promoter regions (Fig. 3). ‘Nagafu No.2’ and ‘Qinguan’ represent the difficult-flowering and easy-flowering varieties, respectively (Xing et al. 2016). Gene expression pattern analysis reveals that, apart from 11 *MdNF-YBs* and 2 *MdNF-YCs* not expressed in either variety, the expression levels of the remaining genes are higher in 'Qinguan'compared to 'Nagafu No.2' (Fig. 4a). This suggests that these gene family members play a crucial role in flowering regulation. Tissue-specific analysis further shows that *MdNF-Ys* are predominantly expressed in leaves, flowers, and fruits across different varieties, highlighting their role in plant reproductive growth.

MdFT1 was previously described as controlling the flowering in apple (Zuo et al. 2024). There are significant differences in *MdFT1* between the two varieties, with *MdFT1* expression in 'Qinguan' being considerably higher than in 'Nagafu No.2' (Fig. 5a). Within 24 h, *MdFT1* expression first increases and then decreases, peaking at 10 o'clock. Interestingly, *MdNF-YB18* displays a similar expression pattern (Fig. 5b), suggesting that the expression of *MdFT1* may be regulated by *MdNF-YB18*. Overexpressing *MdNF-YB18* in *Arabidopsis thaliana* leads to early flowering, indicating its role in promoting flowering (Fig. 5c). Previous research has linked *AtNF-YB2* and *AtNF-YB3* to flowering (Kumimoto 2008), and *MdNF-YB18* is evolutionarily close to *AtNF-YB2/3* (Fig. 2a), suggesting a conserved functional in promoting flowering.

A single NF-Y subunit alone cannot initiate transcription, they always operate as dimers or trimers (Hwang et al. 2016), particularly NF-YB and NF-YC (Kahle et al. 2005; Hackenberg et al. 2012). Yeast two-hybrid screening identified two MdNF-YC members, MdNF-YC3 and MdNF-YC7. Subcellular localization analysis revealed that they all localized in the nucleus (Fig. 6b), indicating their role as nuclear transcription factors. Yeast two-hybrid and BiFC experiments demonstrated that MdNF-YB18 interacts with both MdNF-YC3 and MdNF-YC7 (Fig. 6a and c), confirming that NF-YB and NF-YC can form a complex.

The NF-YB/NF-YC association is essential for NF-YA binding and specific DNA interactions. NF-YA is also compete with other transcription factors, such as CO because of its similar domain to HAP2, to binding to the dimer (Ceribelli et al. 2008; Wenkel et al. 2006). A mutation in HAP3 suppressed mutations in a different region of HAP2 while promoting the assembly of the complex (Xing et al. 1993), highlighting that NF-YB plays the center role within this family. The expressions of both *MdFT1* and *MdNF-YB18* were triggered by the circadian rhythms (Fig. 5b). This study focuses on whether CO, a key regulator of circadian rhythms, influences the expression of *MdFT1* through MdNF-YB18. Yeast two-hybrid assay confirmed that MdCOLs interact with MdNF-YB18 (Fig. 7a), but only MdNF-YB18 bind directly to the promoter of *MdFT1* (Fig. 7b). Dual-luciferase experiments further demonstrated that MdNF-YB18 activates the epression of *MdFT1*, and its interaction with either MdNF-YC3 or MdNF-YC7 enhances this activation (Fig. 7c). Coustry et al. (1996) proposed that MdCOLs might indirectly adjust the expression of *MdFT1* by regulating NF-YB18 in response to the circadian signals. Therefore, we suggest that MdCOLs bind to the dimer complexes of either MdNF-YB18-MdNF-YC3 or MdNF-YB18-MdNF-YC7 to regulate *MdFT1* in apple and promote flowering transition.

This study provides a comprehensive classification of the MdNF-Y gene family in apples and reveals that overexpression of MdNF-YB18 can induce early flowering. *MdNF-YB18* activates the expression of *MdFT1*, with this activation significantly enhanced by interactions with MdNF-YC3 and MdNF-YC7.

Additionally, while MdNF-YB18 can interact with MdCOLs, only MdNF-YB18 binds to the promoter of *MdFT1*. These findings establish a strong foundation for future research on the functional roles of *MdNF-Ys* and offer candidate genes for molecular breeding aimed at studying flower transition and improving yield.

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Author contribution All authors contributed to the study conception and design. Dong Zhang, Yanrong Lv and Xiaoyun Zhang conceived the research idea and plans. Cai Gao analyzed data and prepared the manuscript. Pengyan Wei, Zushu Xie and Pan Zhang performed data mining and bioinformatics. Muhammad Mobeen Tahir, Turgunbayev Kubanychbek Toktonazarovich and Yawen Shen helped to carry out linguistic assistance and experiments. Xiya Zuo, Jiangping Mao, Dong Zhang, Yanrong Lv and Xiaoyun Zhang revised the manuscript. All authors read and approved the final manuscript.

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Data availability All relevant data can be found within the manuscript and its supporting materials.The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval All authors complied with ethical standards and agreed to publish this manuscript.

Consent for publication All authors confirm that the work described has not been published before and not under consideration for publication elsewhere. Written informed consent for publication was obtained from all participants.

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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